Example 6.12

Determine the mean and variance of the random variable X having the following probability distribution.

$X = x$	1	2	3	4	5	6	7	8	9	10
$P(x)$	0.15	0.10	0.10	0.01	0.08	0.01	0.05	0.02	0.28	0.20

Solution:

Mean of the random variable

$$X = E(X) = \sum_x x P_X(x)$$

$= (1 \times 0.15) + (2 \times 0.10) + (3 \times 0.10) + (4 \times 0.01) + (5 \times 0.08) + (6 \times 0.01) + (7 \times 0.05) + (8 \times 0.02) + (9 \times 0.28) + (10 \times 0.20)$

$$E(X) = 6.18$$

$$E\left(X^2\right) = \sum_x x^2 P_X(x)$$

$= (1^2 \times 0.15) + (2^2 \times 0.10) + (3^2 \times 0.10) + (4^2 \times 0.01) + (5^2 \times 0.08) + (6^2 \times 0.01) + (7^2 \times 0.05) + (8^2 \times 0.02) + (9^2 \times 0.28) + (10^2 \times 0.20)$.

$= 50.38$

Variance of the Random Variable

$$X = V(X) = E\left(X^2\right) - \left[E(X)\right]^2$$

$= 50.38 - (6.56)^2$

$= 12.19$

Therefore, the mean and variance of the given discrete distribution are 6.18 and 12.19 respectively.

Example 6.13

Six men and five women apply for an executive position in a small company. Two of the applicants are selected for an interview. Let X denote the number of women in the interview pool. We have found the probability mass function of X.

$X = x$	0	1	2
$P(x)$	$\frac{2}{11}$	$\frac{5}{11}$	$\frac{4}{11}$

How many women do you expect in the interview pool?

Solution:

Expected number of women in the interview pool is

$$E(X) = \sum_x x P_X(x)$$

$$= \left[\left(0 \times \frac{2}{11}\right) + \left(1 \times \frac{5}{11}\right) + \left(2 \times \frac{4}{11}\right)\right]$$

$$= \frac{13}{11}$$

Example 6.14

Determine the mean and variance of a discrete random variable, given its distribution as follows.

$X = x$	1	2	3	4	5	6
$F_x(x)$	$\frac{1}{6}$	$\frac{2}{6}$	$\frac{3}{6}$	$\frac{4}{6}$	$\frac{5}{6}$	1

Solution

From the given data, you first calculate the probability distribution of the random variable. Then using it you calculate mean and variance.

X	$p(x)$
1	$F(1) = \frac{1}{6}$
2	$F(2) - F(1) = \frac{2}{6} - \frac{1}{6} = \frac{1}{6}$
3	$F(3) - F(2) = \frac{3}{6} - \frac{2}{6} = \frac{1}{6}$
4	$F(4) - F(3) = \frac{4}{6} - \frac{3}{6} = \frac{1}{6}$
5	$F(5) - F(4) = \frac{5}{6} - \frac{4}{6} = \frac{1}{6}$
6	$F(6) - F(5) = 1 - \frac{5}{6} = \frac{1}{6}$

The probability mass function is

$X = x$	1	2	3	4	5	6
$P(x)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

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